

2 Covid etc.

Unit-1

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① Error: True Value - Absolute Value

E_a : Absolute Error $|T_v - A_v|$

E_r : Relative Error $\frac{|T_v - A_v|}{T_v} = \frac{E_a}{T_v}$

E_p : Percentage Error $\frac{|T_v - A_v|}{T_v} \times 100 = \frac{E_r \times 100}{\%}$

② Bisection Method:

$x = \frac{a+b}{2}$ where $a = -ve$ or $a = +ve$
 $b = +ve$ or $b = -ve$

③ Regula falsi / Method of false iteration:

$$x = \frac{f(b)a - f(a)b}{f(b) - f(a)}$$

④ Iteration method:

$$x_0 \neq x_1 = x_0 + f'(x_0)$$

$$|\phi(x_0)| \leq 1$$

else

$$x_1 = x_0$$

$\vdots$

$$|x_n - x_{n-1}| \leq \text{allowed error}$$

eg.

Earthquake, flood, Covid etc.

⑤ Newton Raphson

$$x_1 = x_0 - f'(x_0)$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$|x_1 - x_0| \leq \text{allowed error.}$$

⑥ Secant Method:

$$x = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

$$a = b$$

$$b = x$$

$$f(a) = f(b)$$

$$f(b) = f(x)$$

Unit-2

① Matrix Inversion.

$$\begin{aligned} ax_1 + by_1 + cz_1 &= d_1 \\ a_2 x + b_2 y + c_2 z &= d_2 \\ a_3 x + b_3 y + c_3 z &= d_3 \end{aligned}$$

$$A = XB \quad AX = B$$

$$X = A^{-1}B$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj}^o(A)}{|A|}$$

$$\text{Adj}(A) = (\text{cofactor})^T$$

$$\text{Cofactor} = C_{ij} = (-1)^{i+j} [\text{rest of matrix}]$$

② Gauss Elimination:

$$\begin{aligned} a_1 x + b_1 y + c_1 z &= d_1 & \text{--- (I)} \\ a_2 x + b_2 y + c_2 z &= d_2 & \text{--- (II)} \\ a_3 x + b_3 y + c_3 z &= d_3 & \text{--- (III)} \end{aligned}$$

from eq (I) & (II)

$$b_1' y + c_1' z = d_1' \quad \text{--- (IV)}$$

$$b_2' y + c_2' z = d_2' \quad \text{--- (V)}$$

from eq (III)

$$c_3' z = d_3' \quad z = d_3' / c_3'$$

eg.
Earthquake, flood, Covid etc.

Now making the matrix A an upper triangular Matrix:

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

↓

$$a_2 = 0$$

$$a_3 = 0 \quad b_3 = 0$$

Connecting Matrix A & B.

$$\begin{bmatrix} a_1 & b_1 & c_1 & ; & d_1 \\ a_2 & b_2 & c_2 & ; & d_2 \\ a_3 & b_3 & c_3 & ; & d_3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{a_2}{a_1} R_1, \quad R_3 \rightarrow R_3 - \frac{a_3}{a_1} R_1$$

$$\begin{bmatrix} a_1 & b_1 & c_1 & ; & d_1 \\ 0 & b_2' & c_2' & ; & d_2' \\ 0 & b_3' & c_3' & ; & d_3' \end{bmatrix}$$

$$R_3 \Rightarrow R_3 - \frac{b_3'}{b_2'} R_2$$

$$\begin{bmatrix} a_1 & b_1 & c_1 & : & d_1 \\ 0 & b_2'' & c_2'' & : & d_2'' \\ 0 & 0 & c_3'' & : & d_3'' \end{bmatrix}$$

using Backward Substitution

$$\begin{bmatrix} x = \frac{d_1 - c_1 z - b_1 y}{a_1} \\ y = \frac{d_2'' - c_2'' z}{b_2''} \\ z = \frac{d_3''}{c_3''} \end{bmatrix}$$

⇒ Interpolation (for equal intervals)

③ Newton Gregory :

• forward: -

$$y_p = y_0 + P \Delta y_0 + \frac{P(P-1)}{2!} \Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!} \Delta^3 y_0 + \dots$$

• Backward: +

$$y_p = y_0 + P \nabla y_0 + \frac{P(P+1)}{2!} \nabla^2 y_0 + \frac{P(P+1)(P+2)}{3!} \nabla^3 y_0 + \dots$$

earthquake, flood, Covid etc.

④ Central Difference:

Gauss:

- forward: -, -+, -+-, - - -

$$y_p = y_0 + P \Delta y_0 + \frac{P(P-1)}{2} \Delta^2 y_{-1} + \frac{P(P-1)(P+1)}{3!} \Delta^3 y_{-1} + \dots$$

- Backward: +, +-, +-+, - - -

$$y_p = y_0 + P \Delta y_{-1} + \frac{P(P+1)}{2!} \Delta^2 y_{-1} + \frac{P(P+1)(P-1)}{3!} \Delta^3 y_{-2} + \dots$$

⑤ Stirling's

$$y_p = y_0 + \frac{P(\Delta y_0 + \Delta y_{-1})}{2} + \frac{P^2(\Delta^2 y_{-1})}{2!} + \dots$$

$$\frac{P(P+1)(P-1)}{3!} \left[\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right] + \frac{P^2(P+1)(P-1)}{4!} \Delta^4 y_{-2} + \dots$$

$$\frac{1-y_0+y_1}{2}$$

even: $(P-\frac{1}{2})$

$$\Delta y_0 = 3$$

$$\Delta y_1 = 3$$

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(6) Bessel's

$$y_p = \frac{y_0+y_1}{2} + (P-\frac{1}{2})\Delta y_0 + \frac{P(P-1)}{2!} \left[\frac{\Delta^2 y_0 + \Delta^2 y_{-1}}{2} \right] +$$

$$\frac{P(P-1)(P-\frac{1}{2})}{3!} \Delta^3 y_{-1} + \frac{P(P-1)(P+1)(P-2)}{4!} [\Delta^4 y_{-1} + \Delta^4 y_{-2}]$$

$$+ \dots$$

(7) Laplace Everette's Formulae:

$$y_p = (1-P)y_0 + Py_1 + \frac{P(P^2-1)}{3!} \Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!} \Delta^2 y_{-1}$$

$$- \frac{P(P^2-1)(P-2)(P-3)}{5!} \Delta^4 y_{-2} + \frac{P(P^2-1)(P^2-2^2)}{5!} \Delta^4 y_{-1} + \dots$$

[for Unequal & Equal Both]

(8) Lagrange's:

$$y_p = \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} y_0 +$$

$$\frac{(x-x_0)(x-x_2)(x-x_3)\dots(x-x_n)}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)} y_1 + \dots$$

$$+ \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} y_n$$

Eq.

Earth

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⑨ Newton's Divided Difference :

$$y_p = y_0 + (x_0, x_1) [x - x_0] + \frac{[x_0, x_1, x_2]}{(x - x_0)(x - x_1)} [x - x_0][x - x_1] + \dots + \frac{[x_0, x_1, \dots, x_n]}{(x - x_0)(x - x_1) \dots (x - x_{n-1})} [x - x_0][x - x_1] \dots [x - x_{n-1}]$$

x	y	
4	48	
5	100	$\rightarrow \frac{100 - 48}{5 - 4} = 52$
7	294	$\rightarrow \frac{294 - 100}{7 - 5} = 97$
10	900	
11	1210	
13	2028	

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① Using Newton's forward Interpolation:

$$\bullet \frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 + \frac{2P-1}{2} \Delta^2 y_0 + \frac{3P^2-6P+2}{6} \Delta^3 y_0 + \dots \right]$$

$$\bullet \frac{dy}{dx}(x=x_0) : \text{ at } x=x_0 \quad P = \frac{x_0 - x_0}{h} = 0$$

$$= \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right]$$

• 2nd dif derivative:

$$\bullet \frac{d^2 y}{dx^2}(x=x_0) = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \frac{137}{180} \Delta^6 y_0 + \dots \right]$$

$$\bullet \frac{d^2 y}{dx^2} = \frac{1}{h^2} \left[\Delta^2 y_0 + \frac{6P-6}{6} \Delta^3 y_0 + \frac{12P^2-36P+22}{24} \Delta^4 y_0 + \dots \right]$$

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(II) Derivative Using Newton's Backward Interpolation formulae:

$$y_p = y_n + p \nabla y$$

$$\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_n + \frac{2p+1}{2} \nabla^2 y_n + \frac{3p^2+6p+2}{6} \nabla^3 y_n + \frac{4p^3+18p^2+22p+6}{24} \nabla^4 y_n + \dots \right]$$

$\frac{dy}{dx}(x=x_n)$: at $x=x_n$, $p = \frac{x_n - x_n}{h} = 0$

$$\frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \dots \right]$$

$$\frac{d^2 y}{dx^2} = \frac{1}{h^2} \left[\nabla^2 y_n + \frac{6p+6}{6} \nabla^3 y_n + \frac{12p^2+36p+22}{24} \nabla^4 y_n + \dots \right]$$

$$\frac{d^2 y}{dx^2}(x=x_n) = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \dots \right]$$

(iii) Trapezoidal Rule : $n=1$

$$I = \int_{x_0}^{x_1} f(x) dx = \frac{h}{2} [y_0 + \frac{1}{2}(y_1 - y_0)]$$

$$= \frac{h}{2} [2y_0 + y_1 - y_0] = \frac{h}{2} [y_0 + y_1]$$

$$I = \int_{x_0}^{x_n} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

(iv) Simpson's $\frac{1}{3}$ rule : $n=2$

$$I = \frac{h}{3} [(y_0 + y_n) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2})$$

$$+ 4(y_1 + y_3 + y_5 + \dots + y_{n-1})]$$

(v) Simpson's $\frac{3}{8}$ rule : $n=3$

$$I = \frac{3h}{8} [(y_0 + y_n) + 2(y_3 + y_6 + \dots + y_{n-3})$$

$$+ 3(y_1 + y_2 + \dots + y_{n-2} + y_{n-1})]$$

(vi) Boole's Rule: $n=4$

$$I = \frac{2h}{45} \left[7y_0 + 32y_1 + 12y_2 + 32y_3 + 14y_4 \right. \\ \left. + 32y_5 + 12y_6 + 32y_7 + 14y_8 \right. \\ \left. + \dots + 32y_{n-4} + 12y_{n-3} + 32y_{n-2} + 7y_{n-1} \right]$$

(vii) Weddle's Rule: $n=6$

$$I = \frac{3h}{10} \left[y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6 \right. \\ \left. + 6y_7 + \dots \right]$$

(viii) Taylor's Series Method:

$$\frac{dy}{dx} = f(x, y) = y' = y'(x_0)$$

$$\frac{d^2y}{dx^2} = y'' = y''(x_0)$$

⋮

$$\frac{d^ny}{dx^n} = y^n = y^n(x_0)$$

$$y = y_0 + (x-x_0)y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \dots$$

(ix) Picard's Method:

$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx$$

⋮

$$y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx$$

(x) Euler's Method: $h = \frac{x - x_0}{n}$

$$y_1 = y_0 + hf(x_0, y_0)$$

$$y_2 = y_1 + hf(x_1, y_1)$$

⋮

$$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$$

(xi) Modified Euler's Method:

$$y_1 = y_0 + h * f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} * f(x_0, y_0)\right)$$

⋮

$$y_n = y_{n-1} + hf\left(x_{n-1} + \frac{h}{2}, y_{n-1} + \frac{h}{2} * f\left(x_{n-1}, y_{n-1}\right)\right)$$

eg.
Earthquake, flood

(xii) R-k Method (Runge-Kutta)

Order 1: Euler's Method.

$$y_1 = y_0 + k \quad [k = hf(x_0, y_0)]$$

Order 2: Modified Euler's Method.

$$y_1 = y_0 + k$$

$$k = h * f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} * f(x_0, y_0)\right)$$

Order 3: Runge's Method:

$$y_1 = y_0 + k$$

$$k = \frac{1}{6}(k_1 + 4k_2 + k_3), \quad k_1 = hf(x_0, y_0)$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right), \quad k_3 = hf\left(x_0 + h, y_0 + k_1\right)$$

$$k' = hf(x_0 + h, y_0 + k_1)$$

imp
Order 4:

$$y_1 = y_0 + k$$

$$k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + h/2, y_0 + k_1/2)$$

$$k_3 = hf(x_0 + h/2, y_0 + k_2/2)$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

Eq.

Earthquake

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Unit-4

- using Principal of least Square

$$(e_1)^2$$

$$(e_2)^2$$

$\vdots$

$$(e_n)^2$$

$$\sum_{i=1}^n e_i^2 / n = \text{error for curve } (e)$$

- ① Fit a line : degree 1.

$$y = ax + b$$

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

$n = \text{no. of observation}$

- ② Fit a Parabola : degree 2.

$$y = a + bx + cx^2$$

$$\sum y = na + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

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(iii) ~~Fit~~ Polynomial Curve:-

$y = ax^b$
taking log on both side

$$\log y = \log ax^b$$

$\downarrow \quad \quad \downarrow$
 ~~$\log y$~~ ~~$\log$~~

$$\log y = b \log x + \log a$$

~~$\log y$~~

~~y~~

$$\log y = \log ax^b$$

$$\log y = \log a + b \log x$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \downarrow$
 $y \quad \quad A \quad \quad B \quad x$

$$Y = A + Bx$$

$$\sum Y = nA + B \sum x$$

$$\sum xy = A \sum x + B \sum x^2$$

$$b = B$$
$$a = \text{antilog } A$$

Seq.
Earthquake, 11-1

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(iv) Exponential Curve:

$$y = ae^{bx}$$

$$\left[\begin{array}{l} \log_e e = 1 \\ \log_a a = 1 \end{array} \right]$$

taking $\log_e$ both side

$$\log_e y = \log_e a + bx \log_e e$$

$$\log_e y = \log_e a + bx$$

$\downarrow \quad \downarrow \quad \downarrow \downarrow$
 $y \quad A \quad B \ x$

$$y = A + Bx$$

if $\log_{10}$ (Antilog $(\ln)$)
if $\log_e$ (Antilog (10^x))

$$A = \log_e a \rightarrow a = \text{antilog } A$$

$$B = b$$

(v) Correlation And Regression

Correlation Coefficient $\rightarrow r$ (max. value = 1)
min. value = -1

$$r = \frac{\text{Cov}(x, y)}{S.D_x * S.D_y}$$

$$\text{Cov}(x, y) = \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{n}$$

$$S.D_x = \sigma_x = \sqrt{\frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n}}$$

$$S.D_y = \sigma_y = \sqrt{\frac{\sum y^2 - \frac{(\sum y)^2}{n}}{n}}$$

$$r = \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}} \sqrt{\sum y^2 - \frac{(\sum y)^2}{n}}}$$

✓1 Regression

① Linear

Regression Coefficient

b_{xy} [line of x on y]

b_{yx} [line of y on x]

$$b_{xy} = r \cdot \frac{\sigma_x}{\sigma_y}$$

$$b_{xy} = \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}}$$

Eq.

Ex.

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$$b_{yx} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum x^2 - (\sum x)^2}$$

$$b_{yx} * b_{xy} = r \frac{\sum y}{\sum x} * r \frac{\sum x}{\sum y}$$

$$r^2 = b_{yx} * b_{xy}$$

$$r = \sqrt{b_{yx} * b_{xy}}$$

→ line of X on Y

$$x = cy + d$$

$$\bar{x} = \frac{\sum x}{n}$$

$$(x - \bar{x}) = b_{xy} (y - \bar{y})$$

$$\bar{y} = \frac{\sum y}{n}$$

→ line of Y on X .

$$y = ax + b$$

$$(y - \bar{y}) = b_{yx} (x - \bar{x})$$